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ARTICLES

IDENTITY

P. T. GEACH

I AM ARGUING for the thesis that identity is relative. When one says "x is identical with y," this, I hold, is an incomplete expression; it is short for "x is the same A as y," where "A" represents some count noun understood from the context of utterance—or else, it is just a vague expression of a half-formed thought. Frege emphasized that "x is *one*" is an incomplete way of saying "x is one A, a single A," or else has no clear sense; since the connection of the concepts *one* and identity comes out just as much in the German "*ein und dasselbe*" as in the English "one and the same," it has always surprised me that Frege did not similarly maintain the parallel doctrine of relativized identity, which I have just briefly stated. On the contrary, Frege actually enunciated with all vigor a doctrine that identity cannot be relativized: "Identity is a relation given to us in such a specific form that it is inconceivable that various forms of it should occur" (*Grundgesetze*, Vol. II, p. 254).

Absolute identity seems at first sight to be presupposed in the branch of formal logic called identity theory. Classical identity theory may be obtained by adjoining a single schema to ordinary quantification theory (for bound name-variables):

$$\vdash Fa \leftrightarrow \bigvee x(Fx \wedge x = a). \tag{1}$$

Quine in his *Set Theory and its Logic* attributes to Hao Wang (p. 13) the recognition that (1) will serve as a single axiom schema for identity theory. In the vernacular we may intuitively express the content of (1) by saying: Whatever is true of something identical with an object a is true of a, and conversely.

We readily derive from schema (1) the Law of Self-Identity, "$\vdash a = a$." For if we take "$F\xi$" to be "$\xi \neq a$," then schema (1) gives us:

$$\vdash (a \neq a) \leftrightarrow \bigvee x(x \neq a \wedge x = a), \tag{2}$$

which of course yields "$\vdash a = a$." And there are equally easy proofs

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of what has been called (I believe by Quine) the Indiscernibility of Identicals:

$$\vdash Fb \wedge b = a \leftrightarrow Fa \tag{3}$$

and of theorems asserting the symmetry and transitiveness of identity as a relation. The logical system got by adjoining schema (1) to classical quantification theory is a system with a complete proof procedure; moreover, its interpretation is categorical, in the following sense: If we try to introduce *two* two-place predicables, each separately conforming to schema (1), they turn out to coincide in application. (In this paper, as in my book *Reference and Generality*, I use "predicables" as a term for the verbal expressions called "predicates" by other logicians; I reserve the term "predicate" for a predicable actually being used as the main functor in a given proposition.)

In face of these well-known results, it may seem an enterprise worthy of a circle-squarer to challenge the classical theory of identity. All the same, it has an Achilles' heel, as I hope to show. I must first introduce the notion of a *I-predicable*. Consider a theory T with interpreted terms. What are we to take as the substitution-instances of the "F" used in stating schema (1)? It will not be enough to allow replacement of "F" by one-place predicables actually constructible in T. For suppose "$G\xi\eta\zeta$" represents a three-place predicable that can be constructed in T; then if we consider the results of filling up all but one empty place in such a predicable with name-variables, e.g., "$Gvaw$," we shall certainly want the closure of the formula:

$$Gvaw \leftrightarrow \bigvee x(Gvxw \wedge x = a) \tag{4}$$

to count as holding good by schema (1). Similarly for other many-place predicables of T. We shall say that a two-place predicable in a theory T is an I-predicable in T if it satisfies schema (1) for all constructible expressions of T—the formulas just described being here counted as instances of schema (1).

A two-place predicable is an I-predicable, not absolutely, but in relation to a given theory T. But so far this is no ground whatever for asserting the relativity of identity; for this so far amounts to no more than the fact that what an expression signifies is relative

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to the language we are using. For all that I have said so far, it might be true that if a predicable is an I-predicable in a theory T, then what it expresses in that theory is strict, absolute, unqualified identity.

However, if we consider a moment, we see that an I-predicable in a given theory T *need not* express strict, absolute, unqualified identity; it need mean no more than that two objects are indiscernible by the predicables that form the descriptive resources of the theory—the *ideology* of the theory, to borrow Quine's handy term. "For real identity, though," we may wish to say, "we need not bring in the ideology of a definite theory T. For real identity, *whatever* is true of something identical with a is true of a and conversely, regardless of which theory this can be expressed in; and a two-place predicable signifying real identity must be an I-predicable no matter what other predicables occur along with it in the theory." But if we wish to talk this way, we shall soon fall into contradictions; such unrestrained language about "whatever is true of a," not made relative to the definite ideology of a theory T, will land us in such notorious paradoxes as Grelling's and Richard's. If, however, we restrict ourselves to the ideology of a theory T, then, as I said, an I-predicable need not express strict identity, but only indiscernibility within the ideology of T.

Nor is this all. Objects that are indiscernible when we are confined to the ideology of T may perfectly well be discernible in the ideology of a theory T^1 of which T is a fragment, given that T^1 contains more predicables than T. Of course this means that a predicable of T that is an I-predicable in T is no longer an I-predicable with regard to T^1; there is no logical difficulty about this, since the allowable substitutions for the schematic letter "F" in schema (1) will no longer be the same in T^1 as they were in T.

This abstract argument is readily illustrated by a simple example. Let the theory T be a theory about the expressions of a given natural language; let the range of T's quantifiers be *token* expressions of that language; but let the ideology of T be so restricted that in T we cannot give different descriptions for two tokens of the same type-word. Then if a predicable of T, say "$E\xi\eta$" signifies "ξ is equiform with η," it will be an I-predicable in theory T. But now let us add to T's ideology just one predicable

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that discriminates between equiform tokens; in the enlarged theory T^1, we can express something that is true of a token a but not true of an equiform token b; so in T^1 the predicable "$E\xi\eta$" is not an I-predicable any longer.

But now where have we got to? It looked as though satisfaction of schema (1) gave us a necessary and sufficient condition for a predicable's expressing strict identity. But the naive reading of schema (1), by which we should speak quite unrestrictedly of "whatever is true of so-and-so," is inadmissible because of the semantic paradoxes. If instead we consider a predicable that is an I-predicable *relative to a theory* T, then this need only express indiscernibility relative to the ideology of T, and need not express strict unqualified identity. Moreover, an I-predicable of T, while remaining one and the same unambiguous predicable, may be no longer an I-predicable in a wider theory T^1 containing T as a proper part; but we certainly cannot say that this predicable, although it does not change its sense when T is expanded to T^1, expresses strict unqualified identity in T but no longer does so in T^1. We thought we had a criterion for a predicable's expressing strict identity; but the thing has come apart in our hands; and no alternative rigorous criterion that could replace this one has thus far been suggested. I urged initially on intuitive grounds that there just is no such notion as unqualified identity; it now looks as though my intuition was reliable. I might say: the prosecution rests.

Quine, however, can supply a strong argument for the defense of strict identity. His starting-point is this: Whether we interpret a two-place predicable in a theory T as expressing identity goes along with the way we interpret the quantifiers in T. His point is certainly well taken; we may see this from the theoretical possibility of modifying the usual reading of nested quantifiers so as to avoid altogether the use of a special identity sign—a device sketched out in the *Tractatus*, and further developed in recent articles by Hintikka. So we may, and indeed should, consider quantification theory and identity theory as being like two cogwheels that mesh together and move together.

Quine now proposes that if we find an I-predicable in a theory T, we should construe the range of the quantifiers in T as a class of

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objects for which the I-predicable expresses absolute identity, and construe the other predicables of T correspondingly. In a wider system T^1 the range of the quantifiers may on this principle turn out to be different; and then, although each complete sentence of T will survive unchanged in T^1 with the same truth-conditions as before, the *parts* of each sentence will need reconstruing in certain definite ways.

We can grasp Quine's proposal better if we take a concrete example. Let us consider the letters and words in a particular volume on my bookshelves at Leeds. (This example enables us to avoid problems about identity's surviving change; the letters and words in a book are not changeless entities, but for present purposes the changes they undergo may be ignored.) Suppose that the quantifiers of a theory T range over these letters and words, and that the ideology of T does not discriminate between two token-words that are instances of the same type-word. Let "$E\xi\eta$" be an I-predicable for the theory T. Then at first sight we have a choice between two interpretations: we may treat the quantifiers of T as ranging over the token letters and words in my book and read "Exy" as meaning "x is a token equiform with y"; or on the other hand we may treat the quantifiers as ranging over the type-letters and type-words in my book, and read "Exy" as meaning simply "x is identical with y," as expressing absolute identity. Quine would have us accept this second interpretation, and construe other predicables of T accordingly. For example, "— contains two occurrences of 'e'" would not be construed as "—contains two letter 'e's"—for if we follow Quine, we are not allowed *in T* to quantify over token letters like this—but will rather be construed (in the style of his *Mathematical Logic*) as "— contains two initial segments ending with *the* letter 'e'." (I here allow myself to interject a comment on the unholy mixture of Nominalist and Realist ways of talking that we get in a sentence like "The word 'teeth' contains two 'e's." Of course we know what is intended, just as we do when someone says "A donkey is is an animal beginning with 'd'"; but as Aquinas says about the more dubious texts of the Fathers, such remarks are to be charitably construed and on no account imitated.)

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Let us now suppose that we add to the vocabulary of T a predicable or two enabling us to discriminate between different tokens of the same word or letter. In the enlarged system T^1, each full sentence of T will occur with just the same truth-conditions as it had in T; but the import of subordinate expressions in sentences will be radically changed. The quantifiers will now range over token-words of my volume, not over type-words; and a clause "Exy" will now mean not "x is identical with y" but only "x is a token equiform with y." Similarly, if the one-place predicable "F" means "— contains two occurrences of 'e'," this will be spelled out as "— contains two token letter 'e's," and no longer as "— contains two initial segments each of which ends with *the* letter 'e'." Let us now consider the formula:

$$\bigvee x \bigvee y (Fx \wedge Fy \wedge \neg Exy)$$

In T this will be read as: "There are in Geach's volume two non-identical type-words x and y, and x as well as y is such that two of its initial segments end with *the* letter 'e'." In T^1 it will be read as "There are in Geach's volume two non-equiform token-words x and y, and x as well as y contains two letter 'e's." If we think it out, the changes of import for the bits of the formula cancel out, and the truth-condition of the whole remains unchanged.

As you might expect, I have no knock-down logical answer to this: Quine is hardly going to be caught out in a straightforward logical mistake. But I can show, I think, that this way of interpreting quantifiers so as to get an absolute identity involves a sin against a highly intuitive methodological program, clearly enunciated by Quine himself—and obviously Quine himself ought to be worried by this objection. The program is that as our knowledge expands we should unhesitatingly expand our ideology, our stock of predicables, but should be much more wary about altering our ontology, the interpretation of our bound name-variables. I now quote Quine's own clear statement. "But ontology is, pending revision, more clearly in hand than what may be called *ideology* —the question of admissible predicates. . . . That the ontology should be relatively definite, pending revision, is required by the mere presence of quantifiers in the language of science; for quantifiers may be said to have been interpreted and understood only

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in so far as we have settled the range of their variables. And that the fund of predicates should be forever subject to supplementation is implicit in a theorem of metamathematics; for it is known that for any theory, however rich, there are classes which are not the extensions of any of its sentences" (*The Ways of Paradox*, New York, 1966, p. 232).

An admirable aim; but one that we cannot attain by Quine's device of reading strict identity into I-predicables and construing other expressions in a theory correspondingly. Let us consider a theory T, whose quantifiers are assigned a certain range by Quine's method: if T is a rich theory, we may find within T a number of predicables E^1, E^2, E^3, . . ., each of which is an I-predicable in relation to some corresponding sub-theory T^1, T^2, T^3, . . ., and none of which has the same application *in* T as any other. On Quine's view, E^1 will express strict identity in T^1, and E^2 in T^2, and E^3 in T^3, . . .; and the quantifiers of the theories T^1, T^2, T^3, . . ., must accordingly be reconstrued as having different ranges in each case. We wanted to keep our ontology comparatively fixed while allowing changes in our ideology; but now some quite trivial shifts in ideology—the mere omission of some predicables from a theory—will result in quite large additions to our ontology, to the realm our quantifiers are supposed to range over.

Let us go back to the universe of discourse I introduced to you a little while ago—the words in a given volume on my shelves. There are various ways of counting these words: as John Austin remarked, in a rare flash of perceptiveness, type-words and token-words are just two among many ways of counting words. We may, for example, count the dictionary-entry words in a book; in this case "theism" meaning belief in just one God and "theism" meaning addiction to tea will count, although equiform, as two words; but on the other hand in an article on the philosophy of religion the singular "theism," the plural "theisms," and the possessive "theism's," as in "theism's fundamental mistake," will all count as one word. Now we can find a suitably restricted theory for each of the predicables "— is the same token-word as —," "—is a token equiform with —," "— has the same dictionary entry as —," and ever so many other predicables, so that in the appropriate theory the predicable will be an I-predicable; e.g., for the predicable

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"— has the same dictionary entry as —" we need only assume that the theory's descriptive resources are too meagre to distinguish between two words to which the same dictionary entry corresponds. If now we choose to follow Quine, there will be *in rerum natura* ever so many different domains of words, just in one volume on my shelves at Leeds.

This is only one illustration of the way that shifts of ideology lead to an indecent pullulation of entities in our ontology. Take another example. As I remarked years ago when criticizing Quine, there is a certain set of predicables that are true of men but do not discriminate between two men of the same surname. If the ideology of a theory T is restricted to such predicables, the ontology of T calls into being a universe of androids (as science fiction fans say) who differ from men just in this respect, that two different ones cannot share the same surname. I call these androids surmen; a surman is in many ways very much like a man, e.g., he has brains in his skull and a heart in his breast and guts in his belly, for in these respects two men of the same surname do not differ.

The universe now shows itself as a baroque Meinongian structure, which hardly suits Quine's expressed preference for desert landscapes. We surely need Ockham's razor to cut through this tangle; and it is not too hard to see where to make the cut. We reject absolute identity, but admit as many as we need of two-place predicables of the form "— is the same A as —" where "A" is some count noun. The words in my book (say) will constitute just *one* universe of discourse; but different criteria of relative identity, all applying with equal right within this universe, will be given by different two-place predicables, e.g., "— is the same token-word as —," "— is the same type-word as —," "— is the same dictionary-entry word as —." Similarly, we can count the men in Leeds, or, with a different result, the surmen in Leeds; but Leeds does not contain androids as well as men on that account. It is just that we are numbering the inhabitants of Leeds in two different ways; it is just a different predicable—now the predicable "— is the same *man* as —," and now the predicable "— is the same *surman* as —" (and this means, simply, "— is a man with the same surname as —, who is also a man"), —that is serving as our criterion to keep us from counting *the same one* twice over. Our

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ontology is now firmly under control; only our ideology is liable to expand—we cannot tell in advance how many such identity-predicables we are going to need. (We may observe that one theory may contain many such predicables not coinciding in application; though they will not then all be I-predicables in the theory.) But there could not be any objection in principle to an expandable ideology; that is inevitable anyhow—on logical grounds, not just because we happen to live in a surprising world.

I end this paper with an *amende honorable* to the shade of John Locke. For Locke got over the fundamental thesis of this paper pretty clearly in his *Essay*; admittedly he used, here as elsewhere, the murky jargon of "ideas," but it is very easy to transpose his sentences into a new key and talk about count nouns, as I have done. Locke's way of posing the problem of personal identity may be rephrased thus: Even if every man is a person and every person is a man, we still cannot infer that the predicable "— is the same person as —" and "— is the same man as —" coincide in application; this conclusion needs for its support an elaborate argument in the philosophy of mind, not a little logical manipulation in identity theory. Locke with his contempt for formal logic would have been left unimpressed by a formal "proof" that in that case the same man just is the same person and conversely; his reason for rejecting the "proof" would by my lights have been bad, but I think the rejection is right. I am not here saying whether the same man *is* the same person, and vice versa; all I am saying is that Locke was right in not allowing this to be settled easily, even on the supposition that there are no impersonal men and no nonhuman persons. On my own view "Every man is a person" is tantamount to "Every man is the same person as something or other"; and similarly for "Every person is a man" (*Reference and Generality*, p. 191). The bearing of this in mind may mitigate the paradox; for on this analysis the argument Locke would reject becomes inconclusive or circular.

To make this clear, let us suppose that each Leeds man has just one surname. Then each Leeds man is the same surman as himself, by the explanation of "surman"; so each Leeds man is a surman. On the other hand, since "— is the same man as himself," if true of any man, is true of any other man with the same

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surname as he, each Leeds surman is the same man as himself, i.e., is a man. (I have argued, *loc. cit.*, that it does not make sense to ask *which* man he is the same as.) But it does not follow that, in the case supposed, the predicables "— is the same man as —" and "— is the same surman as —" would in Leeds be coincident in application. Nor are there surmen *as well as* men in Leeds; "surmen" and "men" simply give two ways of counting the inhabitants. Just so it *might* be with "man" and "person," as Locke held.

To be sure, Locke also held that there was no special way an individual named by a proper name had to be self-identical. This view must be false if his other view is true; for by keeping on using a proper name, we mean to refer over and over again to the same "thing"; but if this is not an intention to keep on referring to the same *A*, for some definite interpretation of "*A*" as a count noun, then, as lawyers say, the intention is void for uncertainty, and we have not managed to refer at all. Locke's error on this point has been enormously influential, and I do not repent of my repeated criticism of him in print for the error; but I ought in justice to have added that Locke himself shows the fly the way out of Locke's fly-bottle. But then, Locke's *Essay* is like a mail-order catalogue, and you buy what suits you. To switch to a communication-theory metaphor, the trouble is to make out which part is message and which is noise.

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